Junjie Wu

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Education

Tianjin University, College of Intelligence and Computing, *M.Eng. in Electronic Information* Sep. 2023 – Jun. 2026

- **GPA: 3.8/4.0**
- Advisor: Prof. Qilong Wang; Machine Learning and Data Mining Laboratory
- Research area: visual representation learning, progressing from feature modeling of vision foundation models (ICLR'25) to multimodal vision-language pre-training (ICCV'25) and efficient visual token compression for MLLMs (ICMR'26), with recent focus on hallucination in multimodal reasoning

Tianjin University, College of Intelligence and Computing, *B.Eng. in Computer Science and Technology* Sep. 2019 – Jul. 2023

- **GPA: 3.7/4.0, top 15%, recommended admission to graduate school**
- Languages: Mandarin Chinese (native), English (**IELTS 7.0**, CET-6)
- Honors: **Huawei Scholarship** (top 1%), Merit Student of Tianjin University, Outstanding Graduate, Regional Silver Award in the China International "Internet+" Innovation and Entrepreneurship Competition

Research Interests

Multimodal Large Language Models; Visual Representation Learning; Domain-Specific Post Training; Agentic Reasoning

Publications

- **Junjie Wu**, Qilong Wang†, Jiangtao Xie, Pengfei Zhu, and Qinghua Hu. "Asymmetric Factorized Bilinear Operation for Vision Transformer." In *Proceedings of the International Conference on Learning Representations* (**ICLR**), 2025.
- **Junjie Wu***, Jiangtao Xie*, Zhaolin Zhang, Qilong Wang†, Qinghua Hu, Peihua Li, and Sen Xu. "Distribution Alignment-based Language-Image Pre-Training for Domain-Specific Data." In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (**ICCV**), 2025.
- Jiangtao Xie*, Zhaolin Zhang*, Ruiren Zeng*, **Junjie Wu**, Qilong Wang, and Peihua Li†. "Effective Visual Token Aggregation for Improving ViT Classification." *Image and Vision Computing* (**IVC**), 2026.
- Jiangtao Xie*, **Junjie Wu***, Zhaolin Zhang, Qilong Wang, and Peihua Li†. "GDT-VLM: Global Distribution Modeling for Visual Token Compression in Efficient MLLMs." In *Proceedings of the ACM International Conference on Multimedia Retrieval* (**ICMR**), 2026.
- **Junjie Wu*** et al. "VIGIL: Causal Visual Evidence for Reliable Multimodal QA." Manuscript in preparation; arXiv preprint forthcoming, 2026.

*: equal contribution; †: corresponding author

Research Experience

The Hong Kong University of Science and Technology (Guangzhou), Remote Research Assistant Apr. 2026 – Jul. 2026

Reliable Multimodal Reasoning & Agentic Thinking-with-Images, advised by Prof. Yongqi Zhang

- Proposed **VIGIL**, a causal visual-evidence grounding framework built on structural causal models and counterfactual intervention with LPE, CVE, and SVL metrics, attributing MLLM hallucination to insufficient causal dependence of answers on visual evidence; introduced evidence-sufficiency gating that abstains when causal visual evidence is lacking, enabling reliable multimodal QA
- Huawei TIR-Bench Challenge: constructed a closed-loop pipeline of data generation, full-parameter SFT, evaluation feedback, and error analysis for the agentic thinking-with-images benchmark; designed a multi-turn tool-use trajectory co-training scheme with 24,730 training samples (86% with multi-turn tool interaction), yielding consistent gains across model scales: +11.9pp (9B), +6.1pp (4B), and +2.4pp (27B) on Qwen3.5

Tianjin University, Graduate Researcher 2023 – 2026

Machine Learning and Data Mining Laboratory, advised by Prof. Qilong Wang

- Developed higher-order visual representation methods for Vision Transformers, including asymmetric factorized bilinear operations and structured sparse channel mappings, leading to a first-author ICLR 2025 publication
- Studied domain-specific vision-language pre-training through image-text distribution alignment and second-order token statistics, resulting in a co-first-author ICCV 2025 publication
- Investigated efficient visual token aggregation and compression for ViTs and MLLMs, including cross-covariance token pooling and global distribution modeling for visual token compression, leading to a co-first-author ICMR 2026 publication

University of California San Diego (UCSD), Remote Research Assistant Jun. 2023 – Oct. 2023

Collaborated with Prof. Pengtao Xie

- Integrated 842 CELLxGENE datasets with Scanpy/anndata and trained an mRNA gene annotation model on the 10,000 most frequent genes
- Extended scBert with the scFormer attention module; trained an MAE-based chest X-ray foundation model on 10M collected images

Industry Research Experience

Glority, Algorithm Researcher May 2024 – Present

Domain-Specific Multimodal Model Development from Scratch **Yarnpal** and **Calo** Apps (crochet & food domains, $10M+ revenue products)

- **Yarnpal**: independently led algorithm planning and end-to-end delivery from scratch under a low-resource setting; built a dual-source data platform (expert annotation + web crawling) with cleaning and augmentation (background removal, multi-view generation, style transfer, hard-sample synthesis), and devised a decoupled image-to-crochet-DSL workflow
- Established a three-stage training pipeline on Qwen3.5-122B (domain knowledge injection → SFT → GRPO reinforcement learning); engineered an in-house rendering engine and visual evaluation platform for objective assessment, plus an online learning loop forming a data flywheel; the model outperforms major commercial models on internal evaluations and now powers the product's core feature
- **Calo**: independently owned algorithm planning, data construction, and deployment of a domain-specific multimodal model; designed a multi-dimensional data-quality assessment system (image-text alignment, ingredient accuracy) and a non-overlapping two-level label taxonomy, completing 2M+ food-data cleaning, recaptioning, and hard-negative mining
- Fine-tuned Qwen3-VL-8B/30B with GPT-o3 knowledge distillation, chain-of-thought reasoning, and food-database RAG; created an evaluation framework combining LLM-as-Judge, rule-based checks, and human annotation; deployed on a single A100 via vLLM (P90 latency 2.3s), winning A/B tests by 5% over a competitor product and 2% over Gemini-3-Flash, now in production

Multimodal RAG & Agent Framework **Lazyfit** App and customer-service bots across multiple apps

- Architected a four-layer agent pipeline combining language understanding, tag-based retrieval, rule filtering, and RAG search for personalized scheduling
- Implemented retrieval and reranking algorithms for customer-service bots, and applied semantic clustering to user feedback for downstream recommendation analysis

Plant Foundation Model Development **PictureThis** App

- Applied Qwen2.5-VL-7B for recaptioning, semantic expansion, and detail completion to improve supervision quality for downstream multimodal training
- Collected 10M+ plant images for large-scale plant classification and studied plant disease image-text alignment with CLIP-style contrastive learning

Shanghai Qiniu Information Technologies Co., Ltd., Summer Algorithm Intern Jun. 2022 – Sep. 2022

- Built a ship detection and recognition system based on YOLOv5 and related detection architectures, with a Streamlit interface for small-target detection experiments
- Contributed to Qiniu Cloud's content safety review engine by building an efficient NSFW detection baseline for large-scale video streams